Course structure

i. Economics of Digital Goods and Markets

ii. Digital Business Model Innovations

iii. Digital Start-up Management

iv. Ethics and Regulatory Risk

v. Customer-oriented Digital Transformation

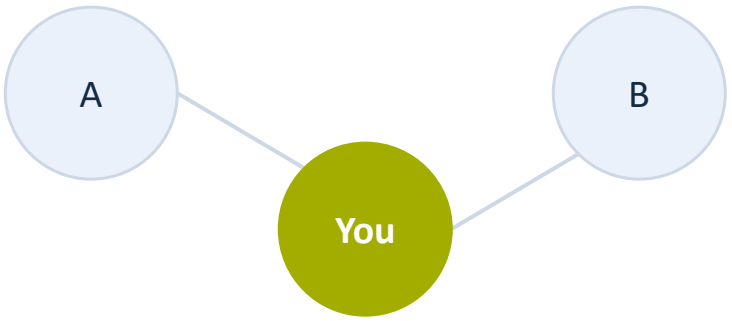
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Direct network effects

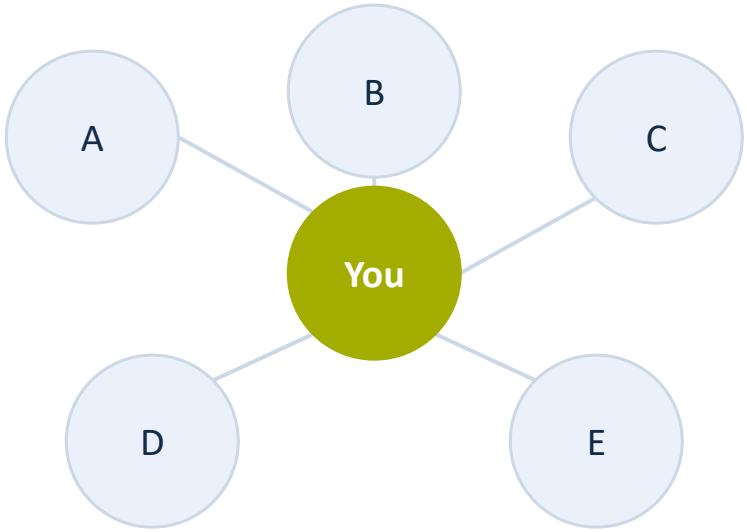
- Direct network effects: the value of a digital platform increases with the number of users in the same user group active on the platform

Example: a messaging app

Smaller network

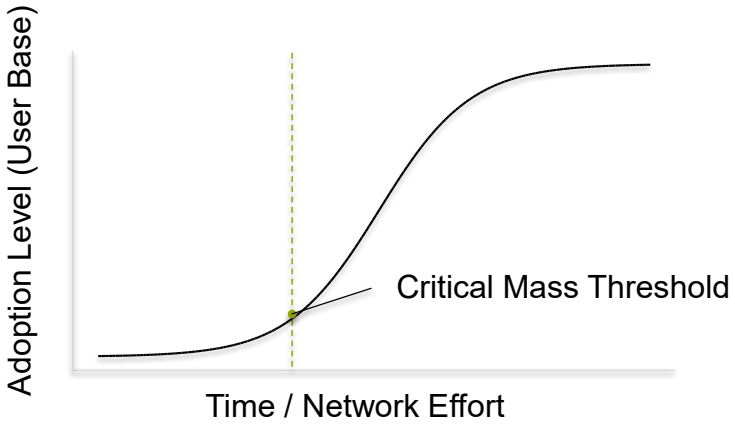


Greater network, greater value



Critical mass for positive direct network effects

- Network effects lead to positive feedback loop once critical mass reached – e.g., new user knows people not yet participating
- Critical mass: the minimum size of a user network at which the value of a platform becomes sufficient to sustain further adoption without continued external incentives
- Threshold depends on how connected participants are and on competing platforms



More participants draw more participants due to network effects



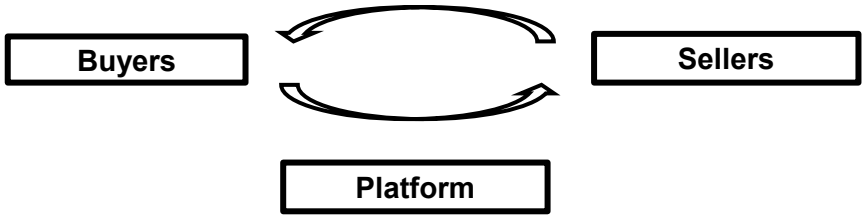
Economides (1996) Network externalities, complementarities, and invitations to enter

Indirect network effects

- Indirect network effects: growth in one user group increases the value of a platform for another user group
- Also referred to as cross-side network effects
- Also lead to feedback loop

Example: an online marketplace

- The more goods, the more buyers browsing the platform
- The more buyers, the more sellers joining the platform to sell their products





Example for both direct and indirect network effects

- The more individuals are on LinkedIn, the more valuable the LinkedIn platform for individuals

- At the same time: The more workers on LinkedIn, the more valuable LinkedIn for companies



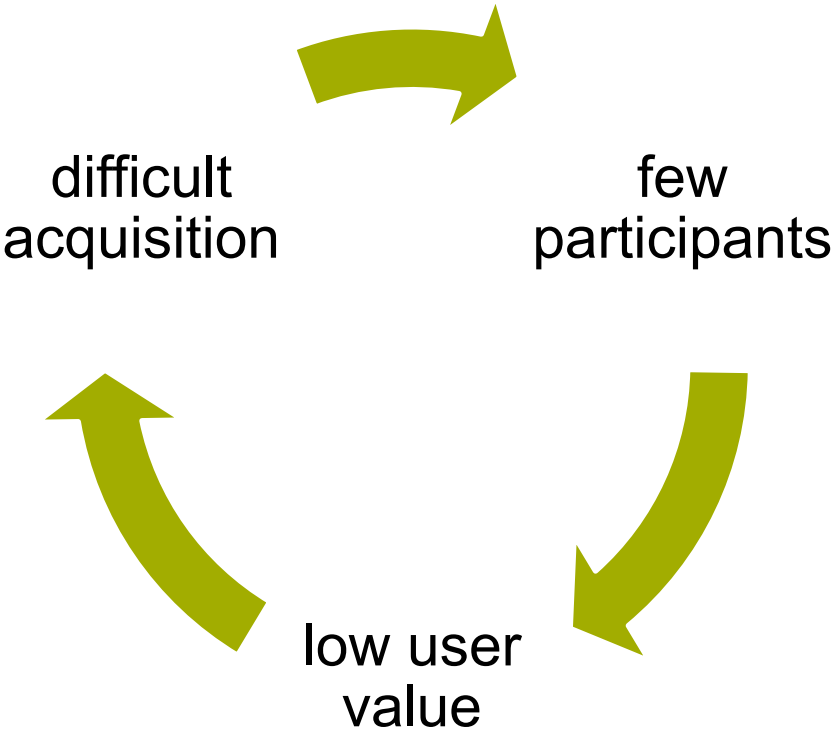
Network effects vs. ordinary economies of scale

	Network Effect	Economies of Scale
Core focus	Demand side	Supply side
Primary driver	More users/connections	Higher production volume
Main benefit	Higher product value	Lower per-unit cost
Typical industry	Software, social media	Manufacturing, logistics
Growth ceiling	Exponential/ compounding	Diminishing returns over time

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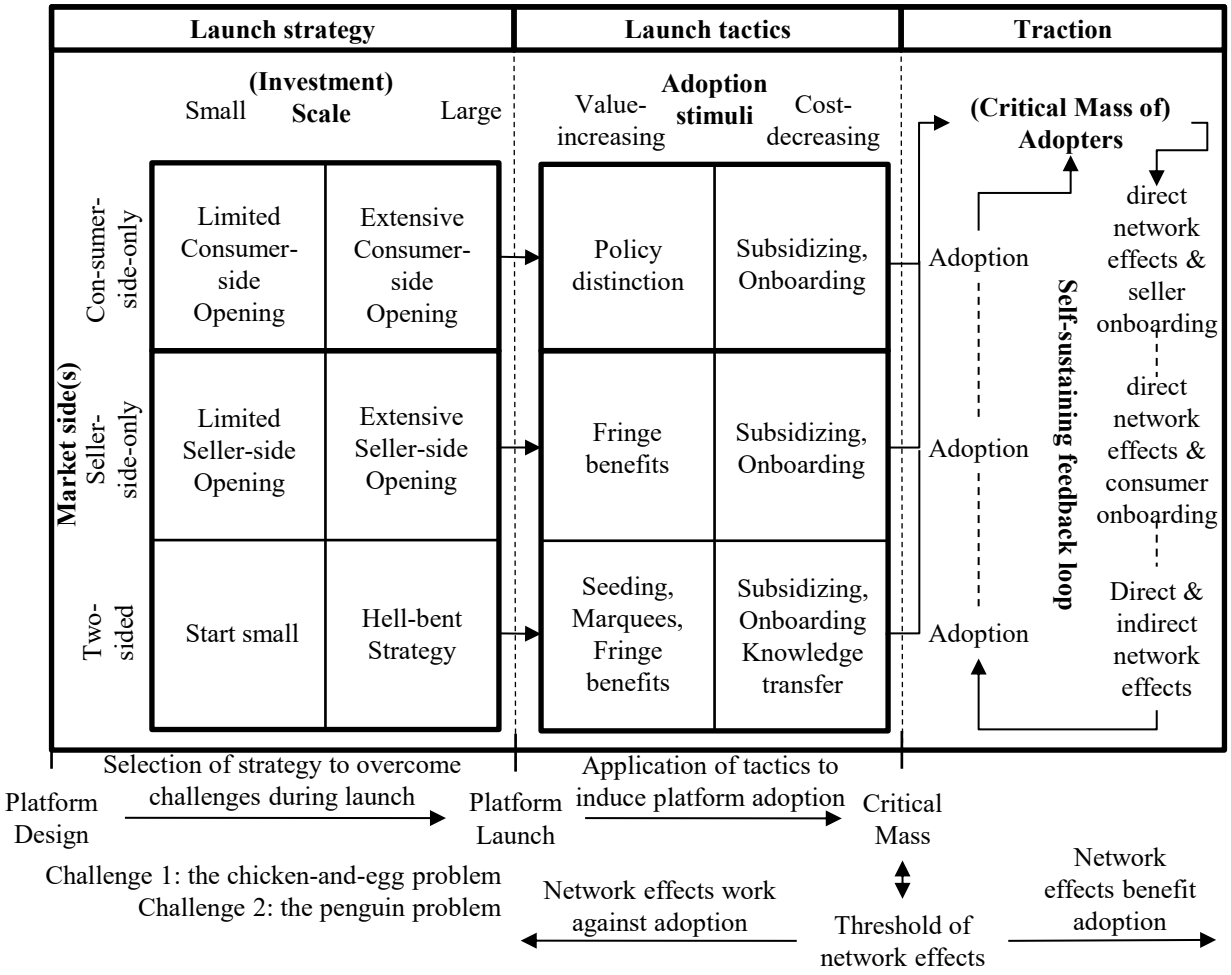
Entrants face a chicken-and-egg problem

- The initial loop



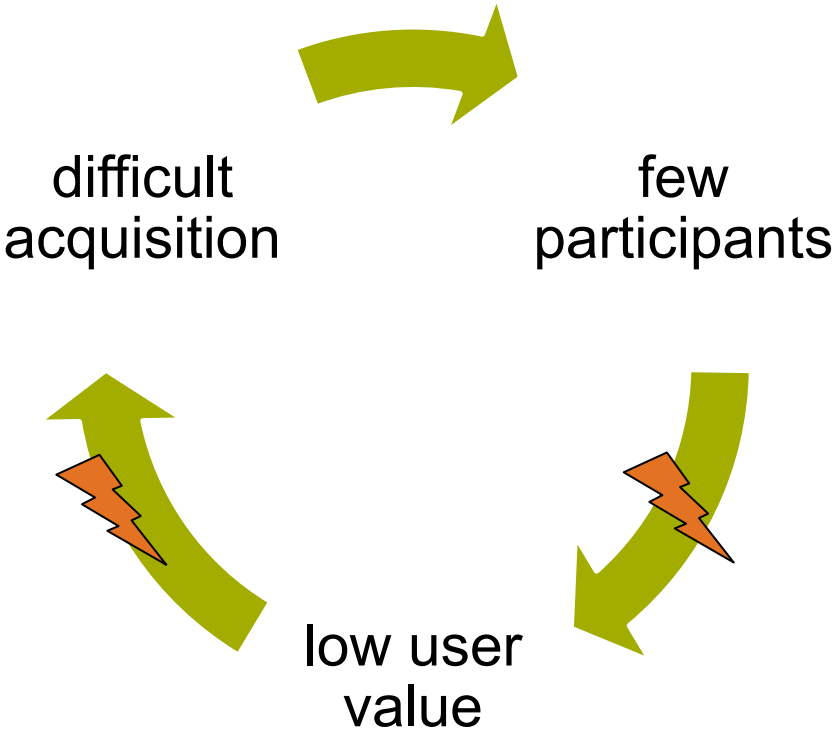
- For a two-sided platform, each side may wait for the other
- A small network may already be useful in a tightly defined niche

Entrants need to make network effects work for them



Start with a dense niche or useful standalone product

- Strategy: Launch in a focused community so the relevant network becomes valuable sooner
- Tactic: Provide useful functionality independent from network

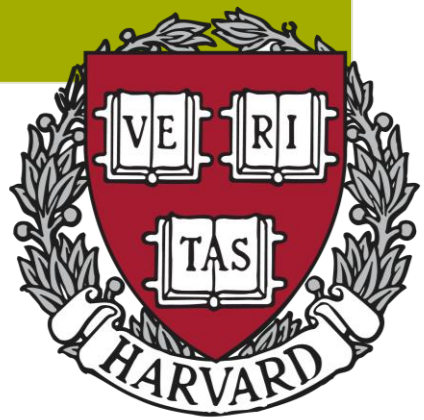


Start small

- Example

Facebook (then called “Facemash”) initially restricted access to Harvard students, then expanded to other universities and eventually the general public

Instead of needing millions of users, Facebook created a dense network within one university first. Critical mass needed is smaller



harvard.edu

Piggybacking on existing platforms

- Example

Airbnb enabled hosts to cross-post their listings on Craigslist, exposing their accommodation offers to Craigslist's established audience of potential travelers

Links in Craigslist listings directed interested travelers to Airbnb, helping the platform attract users and generate bookings without building its entire audience from scratch



Boston Hospitality Review (2016) The Making of Airbnb

Parker & Van Alstyne (2014) Platform Strategy

Launch strategies

Strategy	Description	Examples
Limited consumer-side opening	Launch strategy in which access is initially restricted to a selected group of users, typically through invitations and eligibility criteria. By gradually expanding access, the platform aims to build an engaged, high-quality user base and stimulate network effects	LinkedIn: Initially grew through invitations within professional networks to establish valuable connections and trust Pinterest: Initially used an invitation-only model to gradually build an engaged user community before opening registration to everyone
Limited seller-side opening	Platform launch strategy in which the platform initially focuses on onboarding a limited group of suppliers, often by offering them standalone tools or services that create value independently of consumer participation. Once a sufficient supply base is established, the platform expands its consumer-facing marketplace to activate cross-side network effects	OpenTable: Initially focused on acquiring restaurants by offering reservation-management software. This provided immediate operational value to restaurants, independently of consumer participation Mindbody: Mindbody initially provided management software to yoga and fitness studios. Then, it introduced consumer search functionality and expanded into a consumer-facing booking marketplace
Start small	Platform launch strategy in which a platform initially focuses on a narrowly defined geographic market, community, or customer segment to achieve critical mass and activate network effects within a limited ecosystem before expanding to broader markets	Uber initially launched in San Francisco, concentrating drivers and riders in one city to ensure short waiting times and sufficient ride demand Airbnb: Airbnb initially targeted visitors attending major events with limited hotel availability, such as the 2008 Democratic National Convention in Denver, to concentrate supply and demand

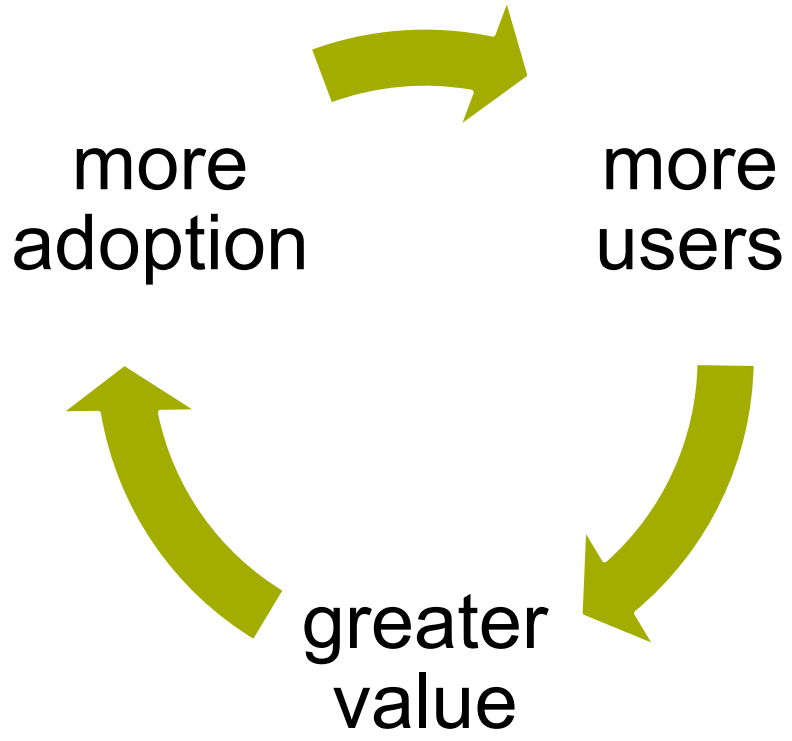
Launch tactics

Tactic	Description	Examples
Onboarding	Actively assist new participants in joining the platform and becoming successful users by reducing setup effort, complexity, and other initial barriers to participation	Airbnb's founders personally visited hosts in New York City and helped improve their property listings by taking professional photographs
Subsidizing	Provide financial incentives or temporarily reduce participation costs for one or both market sides to encourage early adoption and accelerate the development of network effects	PayPal offered cash bonuses to new users for signing up and additional rewards for referring others. These incentives encouraged rapid user acquisition
Fringe benefits	Offer standalone products, services, or functionalities that provide immediate value to participants independently of the platform's network size	Instagram initially offered users easy-to-use photo-editing tools and filters that made smartphone photographs more attractive, even without an established social network
Marquee users	Attract high-profile, influential participants whose presence increases the platform's attractiveness	Twitch has used exclusive agreements, partnerships, and financial incentives to attract and retain prominent content creators
Knowledge transfer	Provide participants with relevant knowledge, training, and best practices to help them use the platform effectively	Etsy provides educational resources, guidance, and seller communities to help merchants improve their shops
Seeding	Platfom creates or supplies content or activities itself to overcome the lack of user-generated contributions	Reddit's founders initially posted links and content themselves, sometimes using multiple accounts, to make the platform appear active

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Positive feedback can favor a leading network

- Growth loop

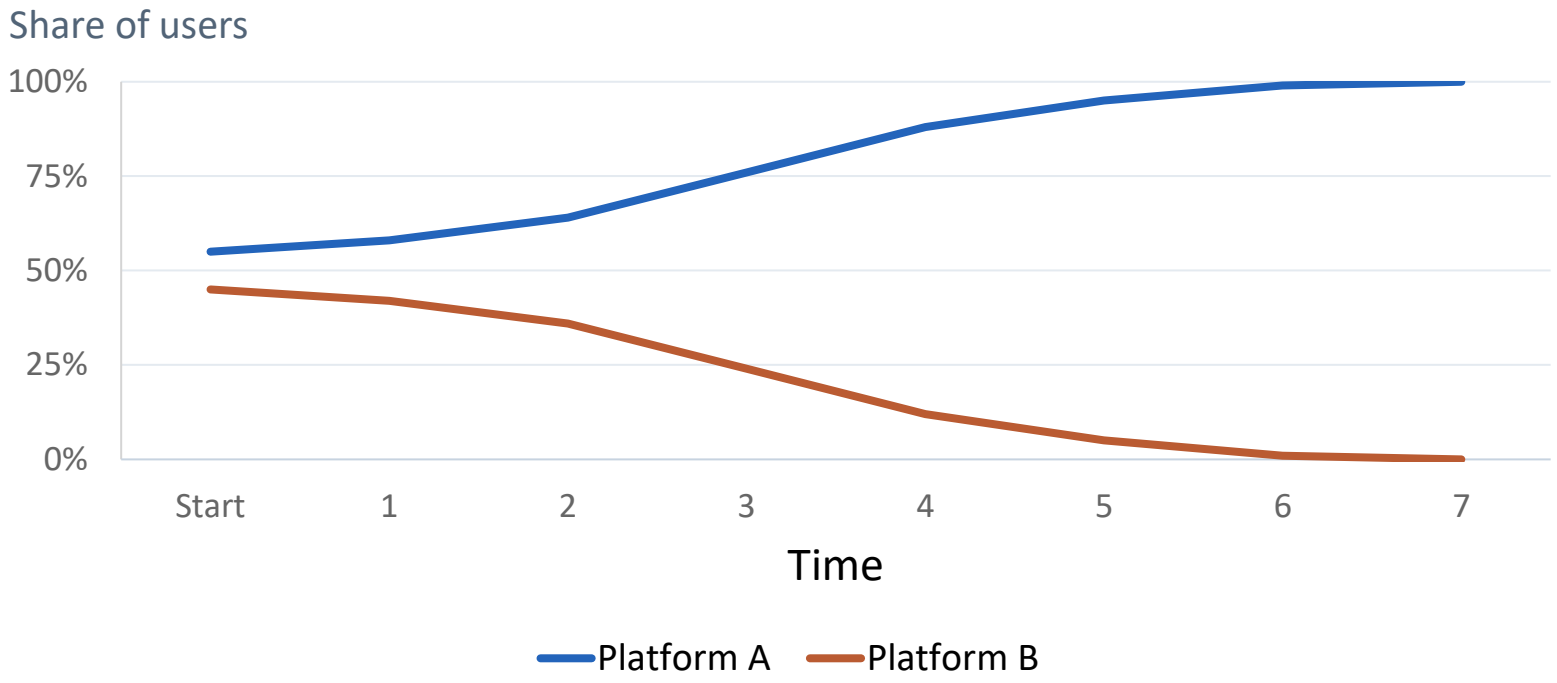


- This may support monetization once the service offers enough value

Network effects may lead to winner-takes-all dynamics

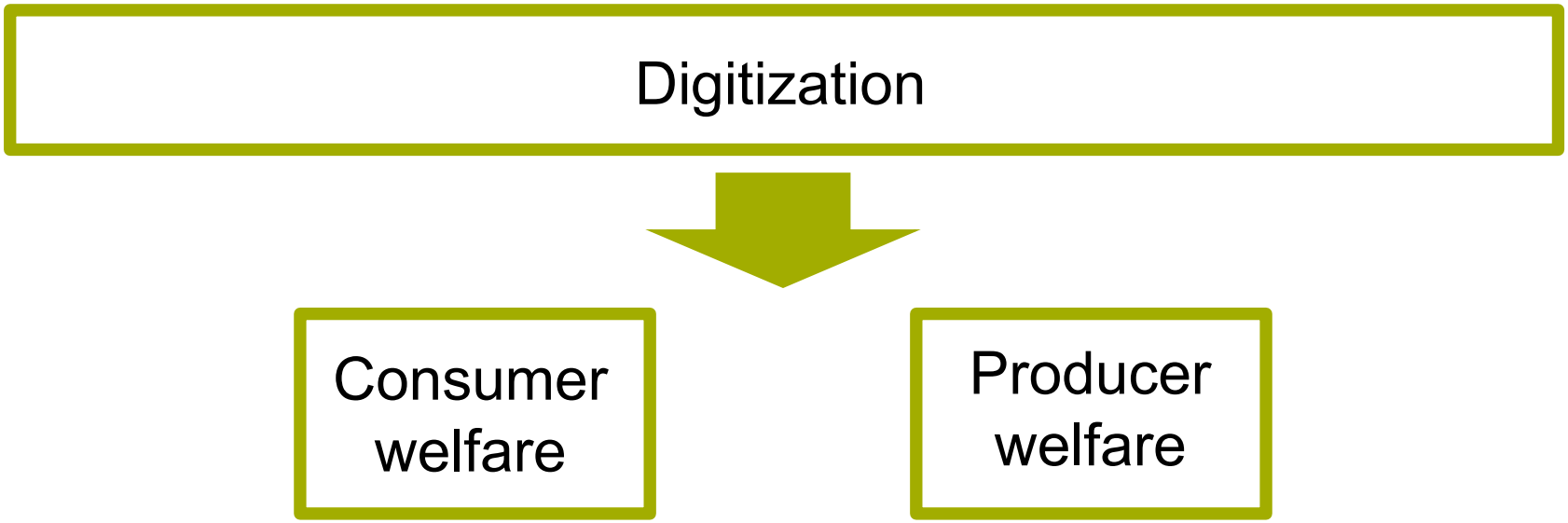
- Markets can tip as adoption increasingly favors larger network

- Network effects do not guarantee complete market tipping



Digitization benefits consumers

How does digitization change consumer welfare and producer welfare?

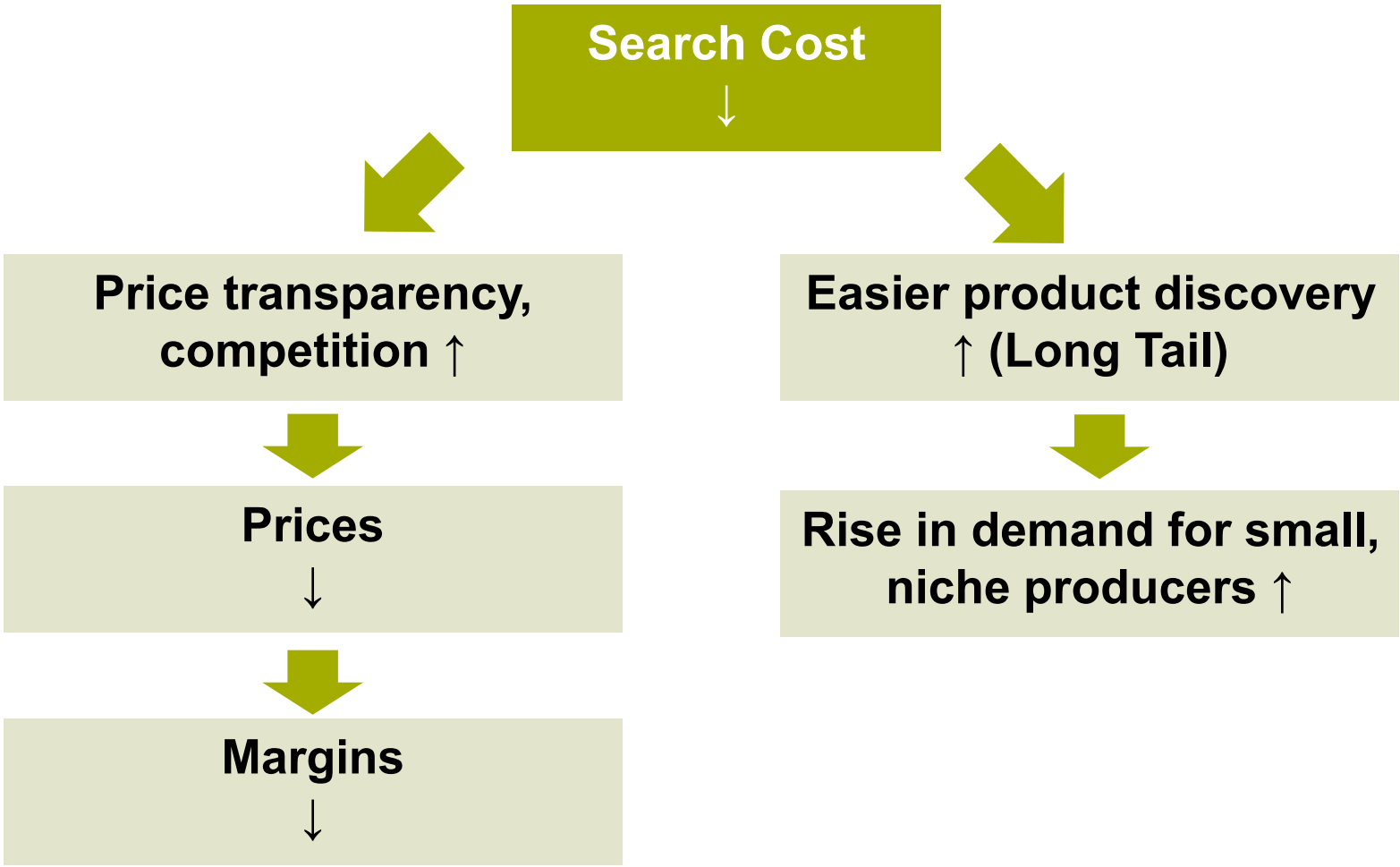


Digitization benefits consumers by reducing costs

	Definition	Example
Search costs	The time, effort and money spent finding and comparing information about products, prices or potential trading partners.	Comparing books on a marketplace platform.
Verification costs	The time, effort and money spent assessing whether information is reliable and whether a person, firm or product is trustworthy.	Using seller ratings and reviews to assess an unfamiliar online seller.
Marginal costs	The additional cost of producing or supplying one more unit of a good or service.	The incremental cost of providing one additional copy of an e-book.

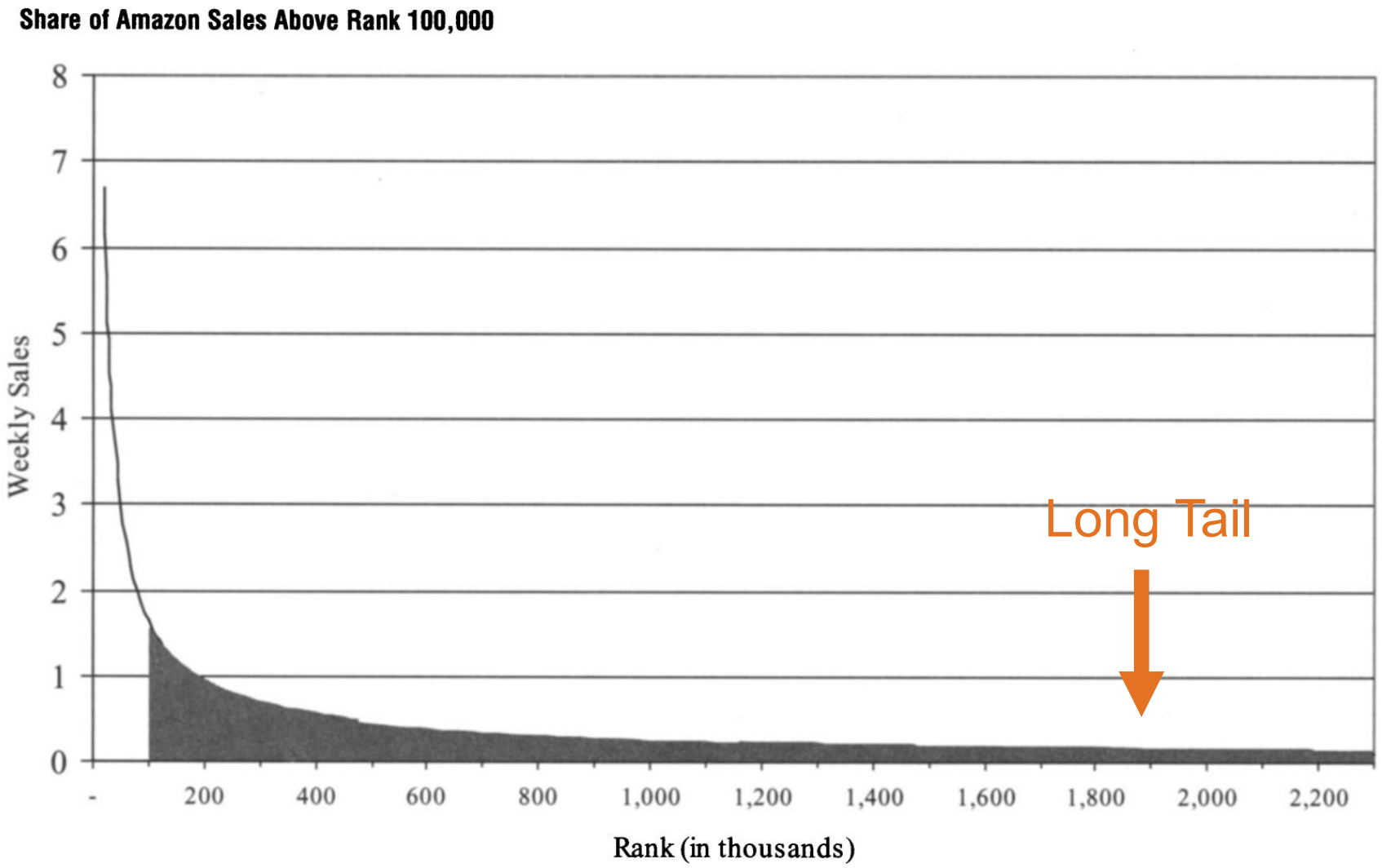
Goldfarb and Tucker (2019) Digital Economics

Margins fall as a result of search cost reduction



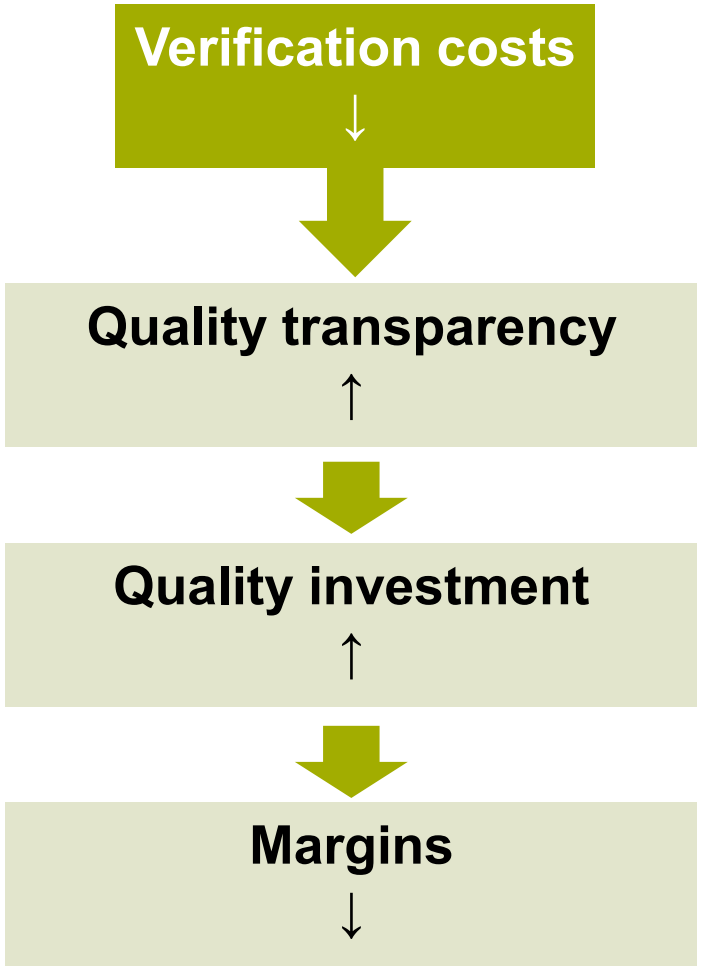
Goldfarb and Tucker (2019) Digital Economics

Better discovery increases product variety



Brynjolfsson et al. (2003) Consumer Surplus in the Digital Economy: Estimating the Value of Increased Product Variety at Online Booksellers

Margins fall as a result of verification costs reduction



Goldfarb & Tucker (2019) Digital Economics

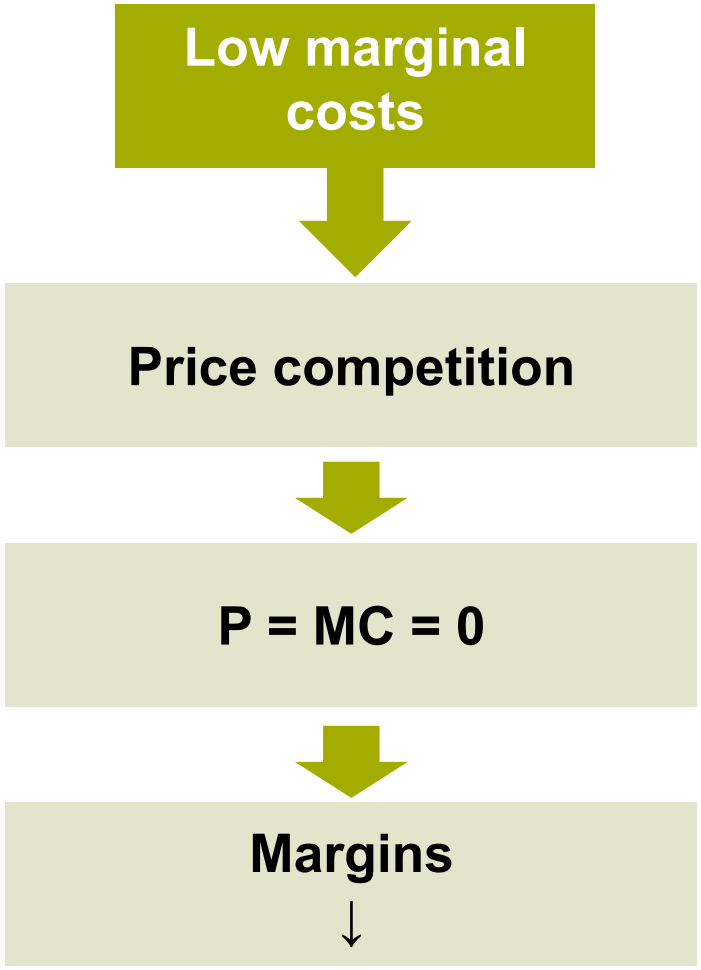
Digital markets typically involve reputation systems

- Reputation signals reduce information asymmetry
- Strengthen incentives for trustworthy seller behavior and providing high quality
- A reputation signal must meet three conditions (Resnick, Zeckhauser et al. 2000)
  - Provide information that allows buyers to distinguish between trustworthy and non-trustworthy sellers
  - Encourage sellers to be trustworthy, and
  - Discourage participation from those who aren't

Before reputational signal	After reputational signal
Buyer knows little about seller	Buyer sees ratings/reviews/history
High verification cost	Lower verification cost
Greater uncertainty about quality	Better-informed quality judgment
Seller behavior has less visible history	Poor behavior can damage future reputation

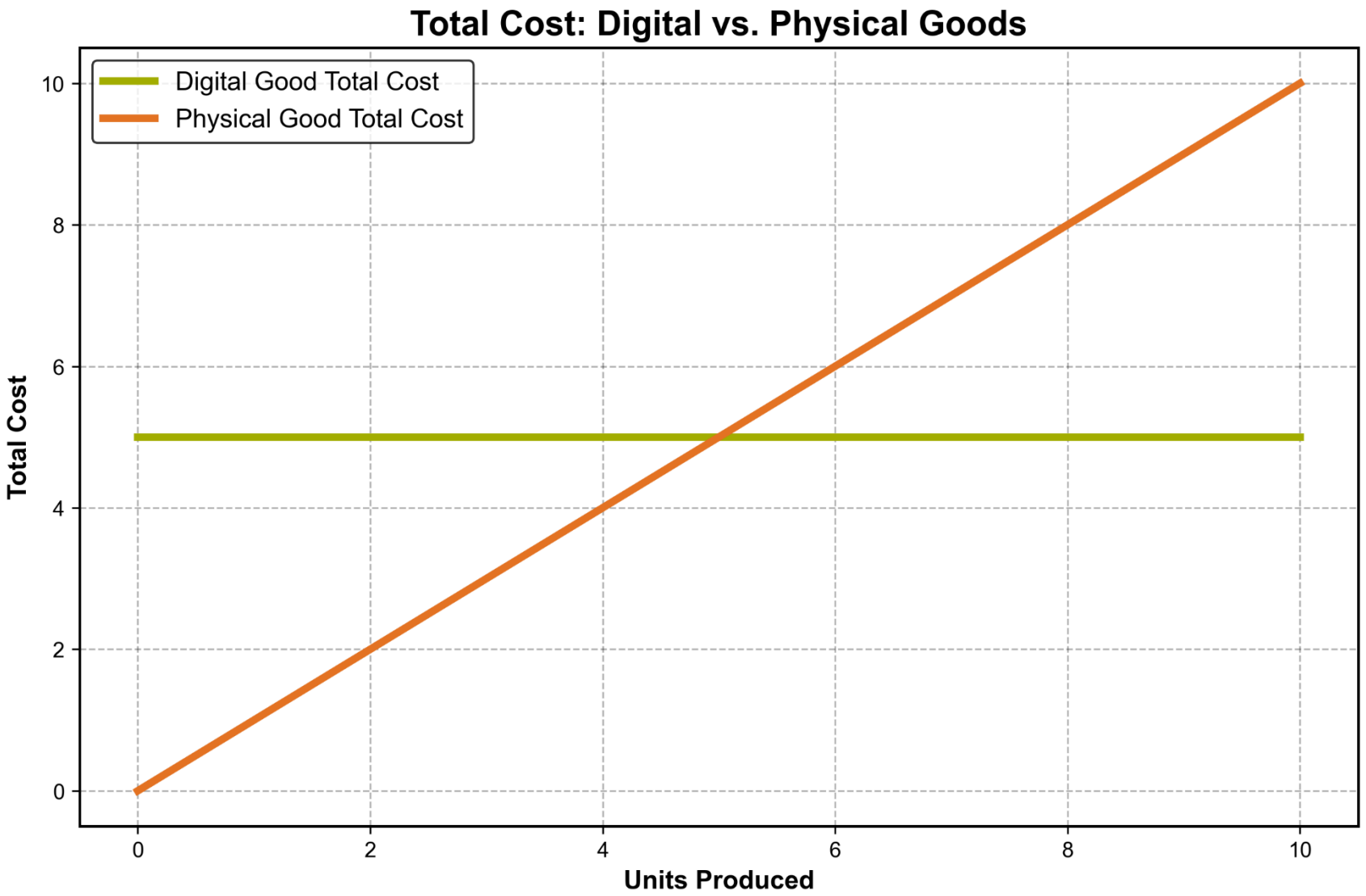
Resnick & Zeckhauser (2002) Trust among strangers in Internet transactions: Empirical analysis of eBay's reputation system
Cabral & Hortaçsu (2010) The dynamics of seller reputation: Evidence from eBay

Prices fall due to low marginal costs and competition



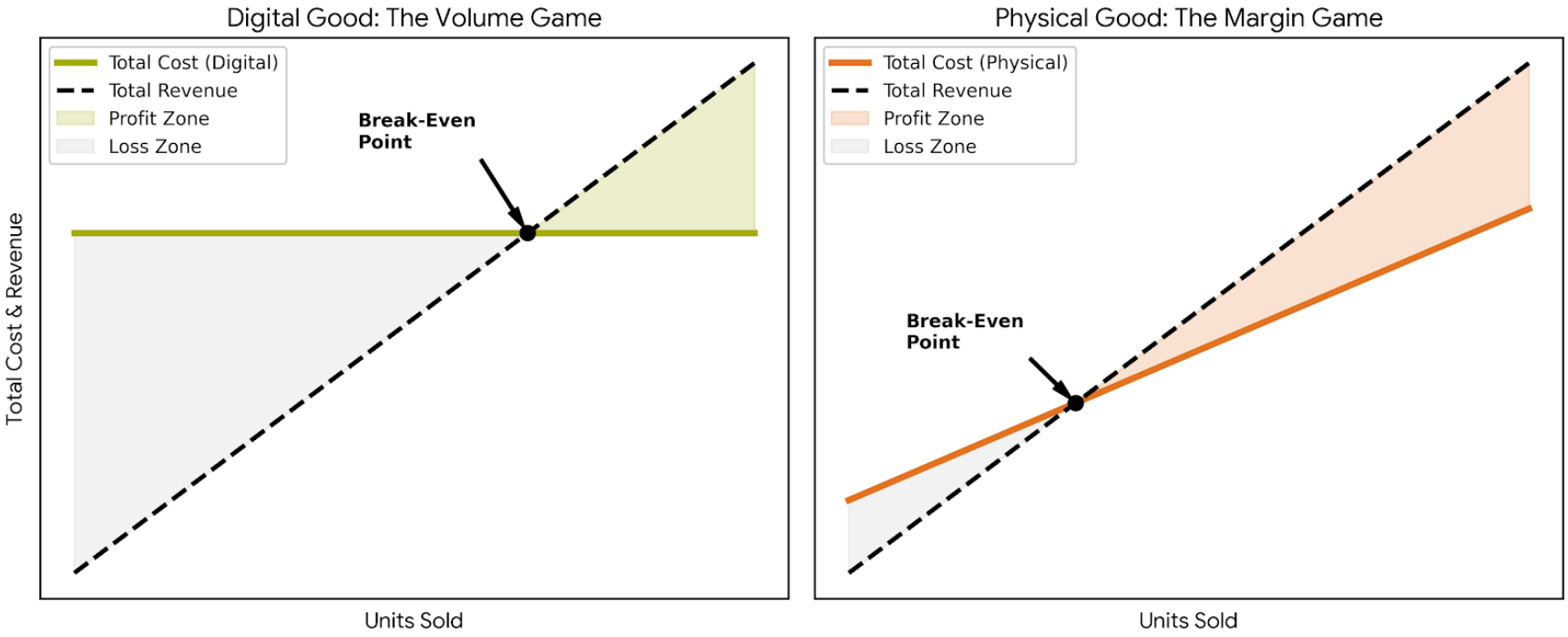
Goldfarb & Tucker (2019) Digital Economics

Digital goods are costly to create and cheap to copy



Stylized illustration; Concept after Shapiro & Varian (1999) and Goldfarb & Tucker (2019)

Low prices require large volume for profits



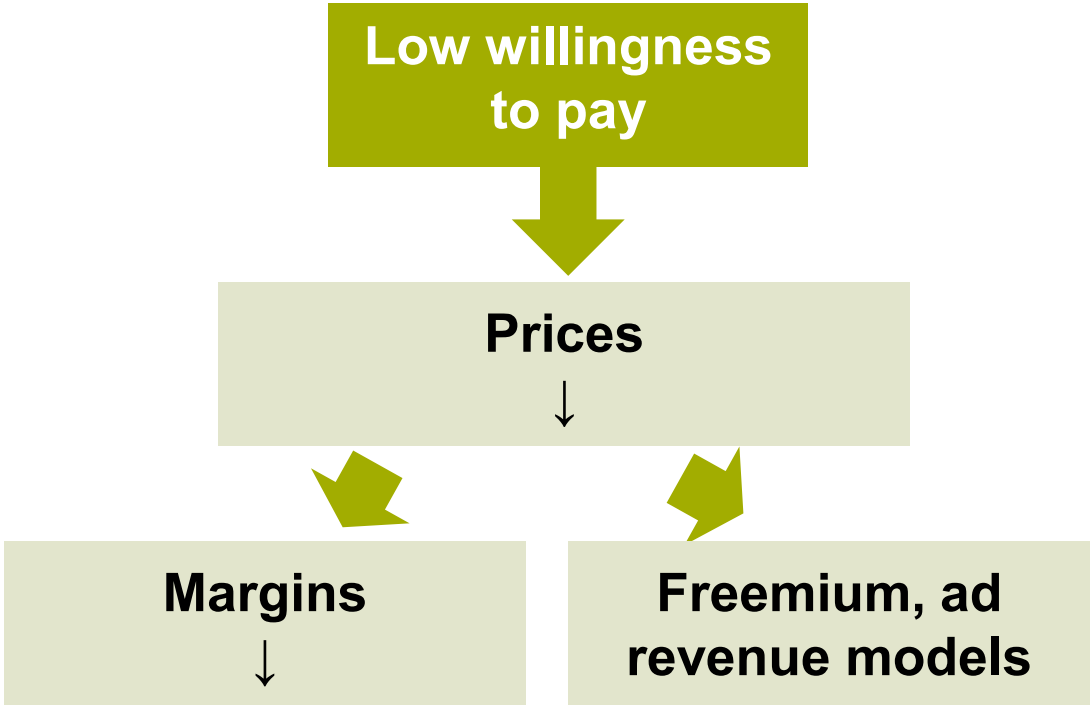
Stylized illustration; Concept after Shapiro & Varian (1999) and Goldfarb & Tucker (2019)

Low marginal costs enables bundling digital goods

- Subscription model at Netflix & Co. is enabled by low MC of digital goods
- Thousands / millions of digital goods
- Different individual preferences average out across the bundle
- Consumers' total valuation of the bundle becomes more predictable, even though they are heterogeneous, allowing to charge one single subscription price and the flate-rate access
- Without bundling: individual prices for each content good, leading to less profit and less value due to absence of flat-rate access

Bakos & Brynjolfsson (1999) Bundling Information Goods: Pricing, Profits, and Efficiency

Prices fall due to low willingness to pay



Goldfarb & Tucker (2019) Digital Economics

Willingness to pay

Willingness to pay

The maximum monetary price a consumer would pay for a specific offer

- Willingness to pay varies across users and products
- Willingness to pay can be low even for high-quality goods

Why is willingness to pay be low?



Consumers value free usage, little value privacy

	Offer A	Offer B
Price	€1.5 per month	€0 per month
Ads	None	Shown in feed and between items
Data use	Only what is needed to run the service; no cross-site tracking; no sale to third parties	Activity and interests used to target ads; Shared with advertising partners
Payment	Money	User: Attention, data Advertiser: compensation for showing ad

Which offer would you choose?

Summary

	Traditional markets	Digital markets
Prices	➢ 0	Often 0 (lower search costs, marginal costs of zero, price competition, low willingness to pay)
Margins	Greater	Lower (quality and price pressure)
Common strategies	Leverage economies of scale, efficient production Quality, price differentiation Building customer relationship	Network effects Facilitate valuable interactions Allowing communication Providing subscription access Informing, entertaining, educating Grow user number and acquire users profitably Monetizing through advertisement